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PI Name:	Hlatky, Lynn Ph.D.		
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PI Email:	Lynn.Hlatky@tufts.edu , LynnHlatky@aol.com	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	(617) 789-2997
Organization Name:	GeneSys Research Institute/Tufts University School of Medicine		
PI Address 1:	736 Cambridge St # CBR-112		
PI Address 2:			
PI Web Page:	http://www.cancer-systems-biology.org		
City:	Boston	**State:**	MA
Zip Code:	02135-2907	**Congressional District:**	8
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
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Key Personnel Changes/Previous PI:			
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Task Description:	Under our NSCOR grant, we made a number of findings that help provide a mechanistic framework for biologically-based risk models. We requested and received a cost extension to finish off experimental characterization of proton and iron irradiation studies to address angiogenesis processes that are important in radiation carcinogenesis and can be viewed as bottlenecks to carcinogenesis advancement. The significance of this in the carcinogenesis risk context lies in the challenge it presents to the standard radiogenic cancer paradigm. This paradigm holds that cancer may be understood as an autonomous expansion of an initiated cell clone, followed by the chance genetic creation of a cancer cell in that clone that then inevitably expands to become a diagnosed cancer. At the basis of these events is thought to be the DNA lesion. We argue that essential population-level and tumor-environmental dynamics can alter this picture dramatically, either promoting cancer development or slowing the course of existing cancers. Their basis extends beyond DNA damage to include the familiar “hallmarks” of cancer that constitute “bottlenecks” to tumor advancement – population-level logistic growth effects, acquisition of angiogenic potential, and competency for invasion, to name three. We have shown that radiation can progress or limit cancers by modifying these bottlenecks. Because these bottlenecks may be significant enough to increase the likelihood of transforming an initiated cell, or to change the likelihood an existing cancer advances to clinical detection over a lifetime, they must be included in our overall assessment of radiogenic cancer risk. It follows that existing models that fail to include such bottleneck effects are inherently flawed in that they do not include all radiation actions bearing on the development of manifest cancer disease. Our successful studies in progression-dependent cancer risk have centered on the functional and molecular characterization and radiation modulation of the angiogenic switch [Abdollahi et al. 2007] and escape from dormancy [Black and Welch 1993, Almog et al. 2009], alone and as modulated by host age. Our goal in the grant extension was to wrap up this work on our panel of primary normal cells and in vivo dormant tumors, in combination with well-characterized human and murine lung tumor models [Hahnfeldt et al. 1999].
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our central tenet is that cancer risk, understood as the likelihood of clinical presentation, is much more than a genetic issue; involving in a critical way effects of host environmental modification e.g. by space radiation. This could be shown by demonstrating that intercellular interactions, as modified by space radiations, can play important, often decisive, roles in how cancer advances towards becoming “manifest disease”. Dramatic support for this possibility comes from an autopsy study that reveals most adults carry frank cancers destined to remain sub-clinical for a lifetime [Black & Welch 1993]. In a followup study, Welch and Black [2010] described the problem of overdiagnosis stemming from improvements in early tumor detection. They found that “very slow” and “non-progressive” tumors that would never become symptomatic constitute an increasingly high percentage of diagnosed disease. For x-ray- or sputum-detected lung tumors, for instance, the excess may be as high as 50%. This would mean a substantial fraction of all detected lung cancers do not pose a symptomatic risk. It thus stands to reason that simple accounting of cells that have been initiated and then transformed to cancer cell status does not accurately estimate the risk for symptomatic disease, or even the risk for cancer detection by routine clinical methods. Bridging the gap, then, becomes a major imperative for properly assessing cancer risk. Indeed, we have shown that space radiations can impact tumor progression in a number of ways, with a trend for one type of space radiation (iron nuclei) to accelerate progression, while for another (protons), the tendency is to forestall tumor development. Implicated for Earth benefit is an exciting new interpretation of potential benefits of proton therapy. Heretofore, it was thought the usefulness of protons for treatment of cancers stemmed only from its physical properties. Unlike gamma electromagnetic radiations which deposit their energy over considerable ranges in the patient, protons are particles that deposit much of their energy at the ends of their tracks. It is thus possible to focus their killing power more precisely to the tumor boundary. However, our work shows the benefit may not stop there. Protons seem to change the way the host interacts with the tumor in such a way that cancer growth may be curtailed, extending patient quality of life and thus the overall benefit of proton therapy.
	Our central tenet is that cancer risk, understood as the likelihood of clinical presentation, is much more than a genetic issue; involving in a critical way effects of host environmental modification e.g. by space radiation. One way to show this is to demonstrate that intercellular interactions, as modified by space radiations, can play important, often decisive, roles in how cancer advances towards becoming “manifest disease”. Our investigations to this end show that cancer-progression-modifying tumor-host interactions indeed occur at a number of levels, including: 1) reduction of angiogenesis by protons, with enhancement by 56Fe (similar to gamma); 2) reduction in the ability of both targeted and non-targeted cells to invade following proton exposures; in contrast with 3) augmentation in ability of tumor and host stromal cells to follow chemotactic stimuli following exposure to 56Fe ions; 4) Reduction in support for tumor proliferation and survival following proton radiation; effects opposite to 5) the stimulation of support for tumor proliferation and survival observed with 56Fe (and gamma). In addition, 6) mouse age generally, as differentially modified by radiation, was found to have a significant effect on cancer progression. Lastly, investigating the potential influence of intercellular interactions within the tumor on radiation modulation of cancer progression, we found that 7) cell-cell interactions among the cancer stem cell (CSC) and non-stem tumor compartments, known to be differentially affected by radiation, can critically affect carcinogenesis course. Suggested is a complex quality-dependent ability of space radiations to influence cancer risks, even after the fact of cancer cell creation. Details: 1) "Reduction of angiogenesis by protons, with enhancement by 56Fe (similar to gamma)": Using pan-genomic arrays, expression of key pro-angiogenic and inflammation-associated genes including VEGF-1A, C-MYC, IL-8, and IL-6, were found generally to be downregulated by protons (1GeV). RT-PCR reconfirmed this down-regulatory trend for a number of specific angiogenesis-regulating genes, including VEGF, IL-8, HIF-1A, and IL-6, c-MYC, FGFR1 and ANGPT1, in human lung microvascular endothelial cells (HLMVEC), normal human dermal fibroblasts (NHCF), and A549 lung carcinoma 6hr after receiving doses of sizes 0.5, 1 and 2Gy. In contrast, when the fibroblast, endothelial and tumor cells were gamma-irradiated at the same doses, the expressions of VEGF, IL-8, HIF-1A and IL-6, were upregulated in a dose-dependent manner relative to control (0Gy) 6hr post-irradiation.

Having established the proton-induced downregulation of these angiogenic genes (6hr post-irradiation), regulation of the corresponding proteins was examined. ELISA assays specific for VEGF and IL-8 were performed after irradiation of endothelial (HLMVEC), fibroblast (NHDF) and tumor (A549) cultures. At 24hr post-irradiation, levels of secreted IL-8 in the culture supernatants were reduced by 31% and 50% in endothelium and tumor cells respectively. Secreted VEGF was also significantly reduced in proton-irradiated fibroblast (17%) and tumor cells (34%) vs sham-irradiated controls (although results with endothelial cells were indeterminate). These findings indicate a dose-dependent downregulation of important angiogenic genes and, for the most part, a significant reduction in protein levels following proton irradiation.

For comparison, 56Fe-induced gene regulation was investigated in the same cell types as for protons. Again, a baseline was obtained by analysis of the iron ion-induced global gene expression signature in the primary human non-transformed cells. Both human lung microvascular endothelial cells (HLMVEC) and normal human dermal fibroblasts (NHDF) were irradiated (0, 0.2, 0.4, and 1Gy) and genome-wide array analysis was performed on RNA isolated 6hr after radiation (or sham) exposure. Expression of key pro-angiogenic and inflammation-associated genes including, VEGF-1A, C-MYC, IL-8, and IL-6, were found to be upregulated. The expression patterns of a number of the angiogenesis-regulating genes, including: VEGF, IL-8, HIF-1A, IL-6, c-MYC, FGFR1 and ANGPT1 were again investigated using quantitative RT-PCR. Results from the quantitative assessment of a number of these genes are shown. Significant upregulation of expression was observed for these genes in both the irradiated HLMVEC and NHDF cells, compared to sham-irradiated controls ($p<0.05$). To further investigate the ability of 56Fe ion irradiation to upregulate specific pro-angiogenic genes, expression levels were also investigated in human lung adenocarcinoma (A549). Just as for primary human fibroblast and endothelial cells, these cancer cells also showed upregulation of VEGF, IL-8, IL-6 and HIF-1A, at 6hr post-irradiation.

2) Reduction in the ability of both targeted and non-targeted cells to invade following proton exposures:

Beyond the suppression of genes critical in angiogenesis, we determined that proton irradiation could also suppress the ability of cells to invade under chemoattractant conditions in vitro. The invasion of the proton-irradiated endothelial, fibroblast, and tumor cells was investigated using a matrigel invasion assay. Compared with untreated controls, proton-irradiated cells (2Gy) showed significantly reduced invasion through the matrigel extracellular matrix. Quantification of the results demonstrated that following proton irradiation, the invasive capacity of A549, NHDF and HLMVEC cells was reduced, by 46, 15 and 14% respectively.

Next, the invasion/migration of endothelial (HLMVEC) in response to soluble factors produced by either proton-irradiated fibroblast (NHDF) or proton-irradiated tumor cells (A549) was analyzed. In an effort to mimic the crosstalk that occurs under in vivo conditions between fibroblast, cancer cells and microvascular endothelial cells, a co-culture transwell chamber system was used. Irradiated or sham-irradiated tumor or fibroblast cells were seeded in the lower chamber. The upper chamber used was an invasion chamber in which untreated endothelial cells were seeded. Since it is known that both fibroblast and cancer cells produce angiogenic factors (e.g. VEGF) that chemoattract endothelial cells, we investigated how proton irradiation would alter this paracrine interaction. Compared to the matched sham-irradiated cell populations, both the 2-Gy proton-irradiated fibroblasts and tumor cells were less effective (19% and 16% respectively) at inducing endothelial cells to invade and traverse to the underside of the membrane. We propose that the reduced invasion of endothelial cells was, in part, a direct consequence of the demonstrated reduced secretion of angiogenic factors from the proton-irradiated cells. Indeed, addition of either recombinant VEGF or IL-8 to the irradiated compartment counteracted the anti-migratory effect of proton irradiation.

3) Augmentation in ability of tumor and host stromal cells to follow chemotactic stimuli following exposure to 56Fe ions:

Extending the finding of increased expression of genes critical in angiogenesis, we determined that iron ion irradiation could also augment the ability of cells to invade under chemoattractant conditions in vitro. Thus far, the invasion of the iron ion-irradiated fibroblast (NHDF) and tumor (A549) cells has been investigated using a matrigel invasion assay. Compared with untreated controls, iron ion-irradiated cells (0.2Gy) showed significantly increased invasion through the matrigel extracellular matrix; even more so than 1Gy. Quantification of the results demonstrated that, following 0.2Gy iron ion irradiation, the invasive capacity of A549 and NHDF cells was increased by 47% and 18% respectively.

We next analyzed the invasion/migration of HLMVEC in response to soluble factors produced by either iron ion-irradiated A549 or iron ion-irradiated NHDF cells. In an effort to mimic the crosstalk which occurs under in vivo conditions between fibroblast, cancer cells and micro-vascular endothelial cells, a co-culture transwell chamber system was used. Irradiated or sham-irradiated A549 or NHDF cells were seeded in the lower chamber. The upper chamber used was an invasion chamber in which untreated HLMVEC cells were seeded. Since it is known that both fibroblast and cancer cells produce angiogenic factors (e.g. VEGF) that are chemoattractant to endothelial cells, we investigated how iron-ion irradiation would alter this paracrine interaction. We found that, compared to the matched sham-irradiated cell populations, both the 0.2-Gy iron-irradiated fibroblast and tumor cells were more effective at inducing endothelial cells to traverse to the underside of the membrane.

We hypothesized in this case that increased secretion of angiogenic factors from the iron ion-irradiated cells was responsible for the increased invasion of endothelial cells. Addition of small molecule inhibitor of VEGF, SU11248 to the irradiated compartment counteracted the pro-migratory effect of iron ion irradiation.

Invasion of tumor (A549), fibroblast (NHDF) and human lung microvascular endothelial cells (HLVMEC) towards conditioned media from 56Fe-exposed cells was also analyzed. A dose-dependent upregulation in invasion was observed. Interestingly, breaking this trend was a dramatic upregulation of invasion of fibroblast cells towards CM from tumor cells irradiated at 0.2Gy, and a downregulation of tumor cells towards CM from LMEC cells irradiated at 0.2Gy. As a matter deserving further investigation, an upward deviation in the vicinity of 0.2Gy from linearity has been consistently noted for the pro-progression effect of 56Fe radiation.

4) Reduction in support for tumor proliferation and survival following proton radiation:

To investigate the modulation of additional aspects of tumor progression by proton radiation, the ability of endothelial (HLMVEC) cells to proliferate was examined in a transwell co-culture assay with proton-irradiated or sham-irradiated tumor (A549) or fibroblast (NHDF) cells. Cancer cells are already known to secrete pro-angiogenic factors e.g. VEGF and IL-8 that can recruit endothelial cells and stimulate their proliferation. Decreased cell proliferation was observed for endothelial cells sharing media with proton-irradiated fibroblasts or proton-irradiated cancer cells, but not for

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endothelial cells sharing media with the sham-irradiated matched controls. In contrast, when this same assay was conducted using photon-irradiated fibroblast or tumor cells, an increased, rather than a decreased, proliferation of endothelial cells was observed.

Next examined was the degree to which proton-irradiated cells (cancer, fibroblast, or endothelial), plated in the upper chambers of transwell plates modulates the proliferation of fibroblast or endothelial cells in the lower chambers. A measurable reduction in proliferation of both fibroblasts and endothelial cells associated with proton-irradiated cancer cells was detected compared to that of controls. We further investigated whether addition of the angiogenic factors VEGF or IL-8 could restore the original proliferation capacity. The proton radiation-induced reduction in responding cell proliferation was partially ameliorated by the addition of VEGF (20ng/ml) or IL-8 (20ng/ml) to the irradiated cell compartment (normalized to sham-irradiated controls). The data suggests that proton-induced suppression of these specific angiogenic factors may well play a direct role in inhibiting adjacent cell proliferation. Such proton-induced suppression of cell proliferation could well have ramifications for treatment optimization. The reduction in proliferation/survival was most pronounced for the NHDF stromal fibroblasts that shared media with 2Gy-irradiated tumor cells. Proliferation levels of the fibroblast cells that shared media through transwells with proton-irradiated tumor cells were less than 50% of their matched fibroblast controls that shared media with sham-irradiated tumor cells. In line with other growth factor rescue experiments, addition of recombinant VEGF or IL-8 partially restored the proliferation levels to 71% and 90% respectively. Similar results were observed for the other cell combinations.

To ascertain the dose dependence of survival of endothelial (HLMVEC), fibroblast (NHDF), or tumor (A549) cells, and whether proton-irradiation induced inhibition of IL-8 might actually contribute to the radiation sensitivity of these cells, clonogenic assays were performed with or without the addition of recombinant human IL-8 (20ng/ml). Cells were irradiated at 1, 2 and 4Gy of protons. It was found that treatment of cells with recombinant IL-8 significantly spared the 2-Gy proton irradiated cells from clonogenic inhibition. The SF2 (survival fraction after 2Gy) of proton and photon/gamma (results not shown) was found to be 0.68 and 0.78 respectively, with the cell killing effect of proton being slightly higher than that of photon radiation.

5) Stimulation of support for tumor proliferation and survival observed with 56Fe (and gamma):

The ability of the HLMVEC cells to proliferate was examined in a transwell co-culture assay with iron ion irradiated or sham-irradiated A549 tumor (or NHDF fibroblast) cells. Contrary to protons, increased cell proliferation was observed for endothelial cells sharing media with iron ion-irradiated fibroblasts or iron ion-irradiated cancer cells, but not for endothelial cells sharing media with the sham-irradiated matched controls.

6) Mouse age generally, as differentially modified by radiation, was found to have a significant effect on cancer progression:

Tumor progression dynamics for young (Y) and middle-aged (MA) mice, with or without iron irradiation at 1GeV/u, were studied for LLC implanted in syngeneic C57BL/6 mice with and without 5x0.2Gy of iron irradiation. Progression was gauged in terms of two quantities: tumor growth rates after approximately exponential growth starts, and “effective” initial tumor volumes, which is proposed to account for differential tumor/stromal “reorganization” times following injection and before growth of the tumor mass.

To do this, the two quantitative aspects of tumor development just mentioned were taken to be the growth rate c and the “effective initial tumor volume” b, obtained by extrapolating the fitted exponential curve backwards to time zero and interpreted as mainly a measure of either stromal incorporation into the tumor, or how supportive the microenvironment was prior to the near exponential growth phase. Growth data were fitted to V = b*exp(ct), with b the geometric mean of the b’s for individual mouse fits, and c the arithmetic mean of the c’s for individual mouse fits.

The data show generally that tumor growth rates are dependent on host age, and that there was a statistically significant decrease in average growth rates c for the unirradiated middle age (MA) as compared to the unirradiated young (Y) hosts, with only a modest increase in the effective initial volume. As regards perturbation by irradiation, the following results were found: i) the effective initial volume b was very significantly decreased by irradiation in both young (Y) and middle-aged (MA) mice; ii) the overall progression decrease by irradiation in the young (Y) mice was due exclusively to a dramatic decrease in effective initial volume; while iii) the more modest decrease in progression by irradiation in middle-aged (MA) mice was due to a slight increase in growth rate compensating for a substantial decrease in effective initial volume; and iv) modest decreases in both average growth rates and effective initial volumes were noted for irradiated middle-aged (MA) mice compared to irradiated young (Y) mice.

To gain some molecular-level understanding, protein-protein interaction (PPI) networks were constructed from the gene array data. Demonstrated were highly regulated protein interactions for tumors of the young (Y) iron-irradiated mice compared with those of the unirradiated groups and of the middle-aged (MA) iron-irradiated compared with those of the unirradiated groups. Overall, the network based on data from tumors growing in irradiated versus unirradiated young (Y) mice demonstrated more regulation of activity than did the network based on data from tumors growing in the irradiated versus unirradiated middle-aged (MA) mice. The general directions of the significantly regulated proteins due to irradiation also coincided with the direction of regulations of genes; mostly down in young (Y) hosts, and a slight bias upward in the middle-aged (MA) hosts. As was also true for the gene regulations, iron irradiation induced a greater genetic response overall in young (Y) compared to middle-aged (MA) hosts.

Interestingly, two central network hubs, FASN (fatty acid synthase) and AKT1 genes, were found to be downregulated within the tumors of young (Y) mice in response to iron-irradiation. FASN is a key metabolic enzyme responsible for lipogenesis and it plays a central role in energy homeostasis. It has been shown that FASN can be highly expressed in aggressive and metastatic tumors. AKT1 is one of three homologous isoforms of the AKT (PKB) protein kinase family that control a wide variety of cellular functions including proliferation, survival, and metabolism. Given the wide range of pro-growth cellular functions AKT1 modulates, it is not surprising it is also implicated in tumor progression and metastases.

Using Ingenuity Pathway Analysis (IPA) software, specific pathways were constructed around FASN and AKT1 that included those genes directly or indirectly up or down regulated for tumor samples from young (Y) or middle-age (MA) mice with and without iron irradiation. Comparisons between groups were done by calculating the log2-fold change, with increasing levels of up- (red) or down- (green) regulation indicated by the darker shades of the respective color and with grey indicating genes present in the network without a 2-fold change under the perturbation studied. In the case of young (Y) unirradiated vs irradiated mice, FASN is seen to be downregulated. By contrast, no FASN regulation was

detected between the tumors from the middle-aged (MA) irradiated and unirradiated groups, consistent with the reduced degree of suppression of progression by irradiation noted for these hosts.

7) Cell-cell interactions among the cancer stem cell (CSC) and non-stem tumor compartments, known to be differentially affected by radiation, can critically affect carcinogenesis course.

In recent years cancer stem cells (CSCs) have been hypothesized to comprise only a minor subpopulation in solid tumors that drives tumor initiation, progression, and metastasis; the so-called "cancer stem cell hypothesis." While a seemingly trivial statement about numbers, much is put at stake. If true, the conclusions of many studies of cancer cell populations could be challenged, as the bulk assay methods upon which they depend have, by, and large, taken for granted the notion that a "typical" cell of the population possesses the attributes of a cell capable of perpetuating the cancer, i.e., a CSC. In support of the CSC hypothesis, populations enriched for so-called "tumor-initiating" cells have demonstrated a corresponding increase in tumorigenicity as measured by dilution assay, although estimates have varied widely as to what the fractional contribution of tumor-initiating cells is in any given population. Some have taken this variability to suggest the CSC fraction may be nearly 100% after all, countering the CSC hypothesis, and that there are simply assay-dependent error rates in our ability to "reconfirm" CSC status at the cell level. To explore this controversy more quantitatively, we developed a simple cellular automaton model of CSC-driven tumor growth dynamics. Assuming CSC and non-stem cancer cells (CC) subpopulations coexist to some degree, we evaluated the impact of an environmentally dependent CSC symmetric division probability and a CC proliferation capacity on tumor progression and morphology. Our model predicts, as expected, that the frequency of CSC divisions that are symmetric highly influences the frequency of CSCs in the population, but goes on to predict the two frequencies can be widely divergent, and that spatial constraints will tend to increase the CSC fraction over time. Further, tumor progression times show a marked dependence on both the frequency of CSC divisions that are symmetric and on the proliferation capacities of CC. Together, these findings can explain, within the CSC hypothesis, the widely varying measures of stem cell fractions observed. In particular, although the CSC fraction is influenced by the (environmentally modifiable) CSC symmetric division probability, with the former converging to unity as the latter nears 100%, the CSC fraction becomes quite small even for symmetric division probabilities modestly lower than 100%. In the latter case, the tumor exhibits a clustered morphology and the CSC fraction steadily increases with time; more so on both counts when the death rate of CCs is higher. Such variations in CSC fraction and morphology are not only consistent with the CSC hypothesis, but lend support to it as one expected byproduct of the dynamical interactions that are predicted to take place among a relatively small CSC population, its CC counterpart, and the host compartment over time.

Overall conclusions:

Cell-cell interactions between the tumor and host, and between tumor cell subcompartments, affect radiation-induced cancer progression, and thus cancer risk, in a highly radiation-quality-dependent fashion. We focused on two radiation types, 56Fe and protons, and found them to generally encourage and discourage, respectively, a number of molecular and functional indicators of progression. This fact must be considered alongside the risks of cancer cell creation when making conclusions as to astronaut safety during long-term space flight. Closer to home, the favorable tumor-host dynamical effects noted for proton radiation in space may work to supplement its well-established physical attributes as a therapeutic modality.

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